

Subject Description Form

Subject Code	SFT5102
Subject Title	Life Cycle Assessment of Fashion Products
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The subject aims to provide students an insightful and holistic understanding of the life cycle assessment and apply it on fashion products. It also encourages students to explore creative approaches for a big change in the lifecycle of fashion products.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Know the general principle of life cycle assessment b. Implement life cycle assessment on specific fashion products c. Organize report on assessment results d. Provide suggestions on fashion product sustainability e. Catch the progress & trend in life cycle assessment
Subject Synopsis/ Indicative Syllabus	1. Life Cycle Principle of Specific Fashion Products Typical fashion products are involved; the general principle is disseminated into key points to implement. 2. Environmental impact assessment of the product It is the core task to be completed step by step: -- Material phase impact -- Manufacturing phase impact -- Retail phase impact -- Use phase impact -- Disposal phase impact 3. Social impact assessment of the product The product enrichments to workers, community, supply chain and society. 4. Economic impact assessment of the product The product contribution to brand reputation, company image, and local economy. 5. Company initiatives & institute researches Inspiring cases of famous products, innovative research, and leading brands. 6. Certificate practices in life cycle assessment Popular certificates in different counties are followed to guarantee the positive result of life cycle assessment.
Teaching/Learning Methodology	Fundamental concepts and logic approaches will be introduced through lectures for analyzing and resolving problems. Group discussions within tutorial classes will allow students to practice approaches of assessment and negotiation. Assignment will be given to students to stimulate self-learning, paying attention to wide areas and update advances in fashion

	sustainability assessment.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Coursework presentation	50%	✓	✓	✓	✓		
	2. Essay assignment	50%	✓	✓	✓	✓	✓	
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The coursework presentation allows students to work in groups and discuss to complete a problem resolving project with learnt life cycle assessment principle and approaches, organize collected materials and present their ideas on practical pathway leading to better sustainability fashion products. Essay assignment requires students think individually, collect references, resolve problems with learnt knowledge, and organize materials to demonstrate their creative initiatives for solidifying life cycle assessment on fashion products.							
Student Study Effort Expected	Class contact:							
	▪ Lecture		26 Hrs.					
	▪ Tutorial		13 Hrs.					
	Other student study effort:							
	▪ Essay assignment		40 Hrs.					
	▪ Project preparation		30 Hrs.					
	Total student study effort		109 Hrs.					
Reading List and References	Books: Blackburn, R. S. (2009). <i>Sustainable textiles: Life cycle and environmental impact</i>. Woodhead Publishing limited. Ralph Horne, Tim Grant, Karli Verghese (2009). <i>Life cycle assessment: principles, practice, and prospects</i>. Collingwood, Vic. CSIRO Pub. Subramanian Senthilkannan Muthu (2015). <i>Handbook of Life Cycle Assessment (LCA) of Textiles and Clothing</i>. Elsevier Ltd. Woodhead Publishing limited.							
	Journal papers:							
	Felix M. Piontek, Martin Müller. Literature reviews: Life Cycle Assessment in the context of Product-Service Systems and the Textile Industry. <i>Procedia CIRP</i> 2018, 69, 758 – 763.							

	Bahareh Zamani, Gustav Sandin, Greg M. Peters. Life cycle assessment of clothing libraries: can collaborative consumption reduce the environmental impact of fast fashion? <i>Journal of Cleaner Production</i> 2017, 162, 1368-1375. Thesis: B.R. de Haan. Exploring life cycle sustainability in the fashion industry. Master's Thesis for the Environment and Society Studies programme, Nijmegen School of Management, Radboud University, June, 2017.
--	---